

INTRODUCTION TO AI AND APPLICATIONS

Module 3

Module 3: Prompt Engineering Techniques for ChatGPT

1. Introduction to Prompt Engineering Techniques

- Prompt engineering involves crafting specific instructions (prompts) to guide how an AI model generates responses.
- By providing clear and structured prompts, developers can direct the AI's output to be relevant, accurate, and high-quality for a wide range of tasks.
- **Goal:**
 - The main goal of prompt engineering is to shape the model's behavior, ensuring that the responses meet the desired standards and objectives for a given task
- **Key Techniques in Prompt Engineering**
 - **Standard Prompts:** Simple prompts that directly ask the AI to perform a specific task.
 - **Role Prompting:** The model is given a specific role to play, influencing the tone, style, or behavior of the response.
 - **Seed-Word Prompting:** The prompt provides a seed word or phrase that the model should build upon, guiding its response based on that initial cue.
 - **Conditional Prompting:** Conditions or limitations are set for the model, such as "If this, then that".

Module 3: Prompt Engineering Techniques for ChatGPT

1. Introduction to Prompt Engineering Techniques

- **Why is Prompt Engineering Important?**
 - **Relevance:** Well-crafted prompts ensure the model produces relevant responses, making them more aligned with the task's objectives.
 - **Accuracy:** Prompt engineering helps the model generate responses that are factually correct and appropriate for the given context.
 - **Versatility:** Different types of prompts can be used to adapt the model's output to fit a wide variety of tasks, from casual conversations to formal reports.
- **The Future of Prompt Engineering**
 - As AI continues to advance, prompt engineering will become increasingly important for optimizing language models. Future developments may involve:
 - Automated prompt generation to streamline the process.
 - Dynamic prompts that adjust based on user interactions or context.
 - Contextual awareness in prompts, allowing the AI to respond more naturally and flexibly.

Module 3: Prompt Engineering Techniques for ChatGPT



2. Instructions Prompt Technique

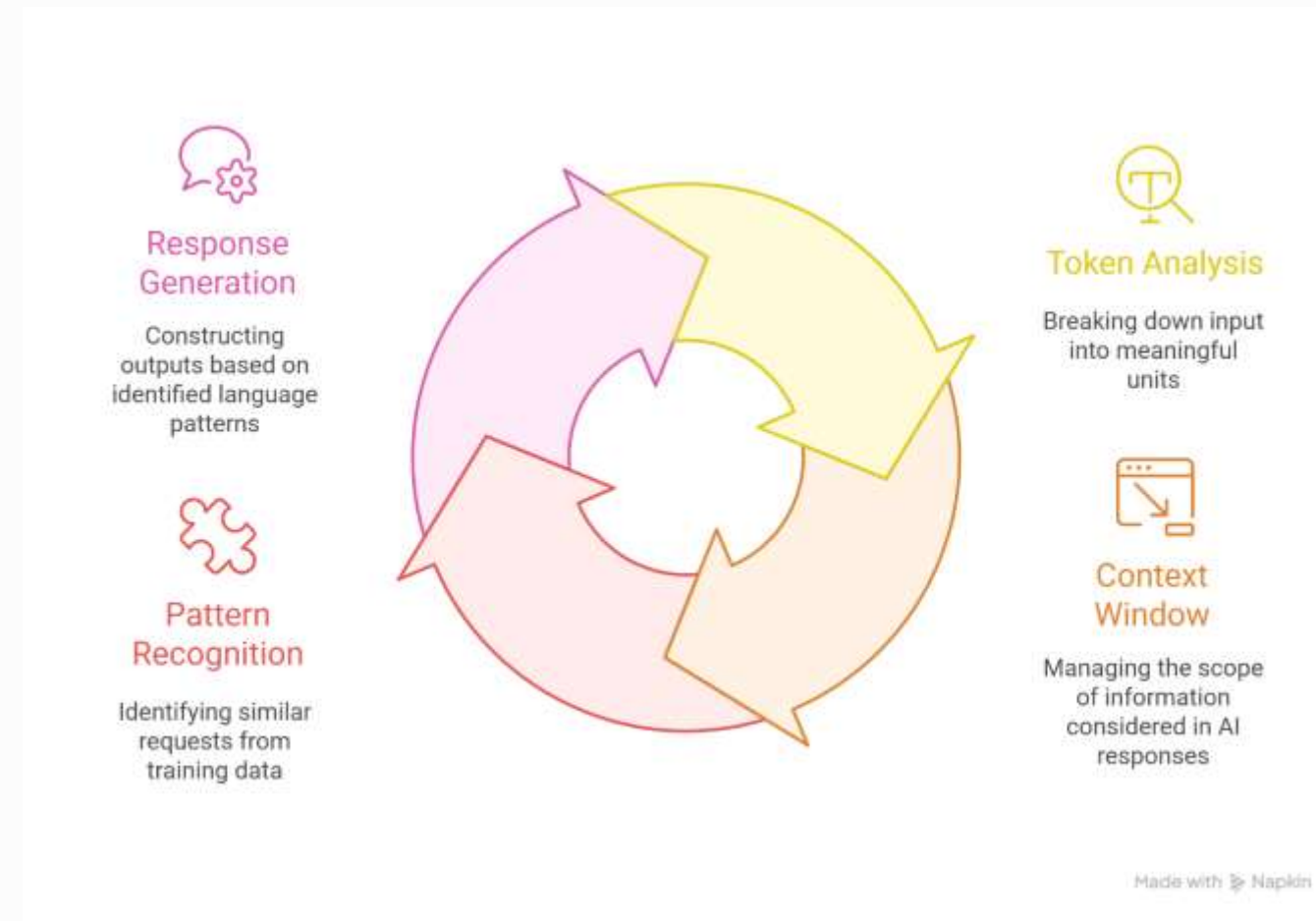
- The Instructions Prompt Technique is a method used to guide the output of natural language processing models like ChatGPT by providing specific instructions that the model must follow. This ensures the generated text aligns with the desired objectives and requirements of the task.
- **Purpose:** Directs the model to produce outputs based on specific guidelines or criteria.
- **Customization:** Allows for tailoring the model's response to meet particular goals, such as professionalism, accuracy, or legal compliance.
- **How It Works:**
 - **Step 1:** Provide clear instructions to the model. These instructions specify how the model should approach the task or the style of the output.
 - **Step 2:** The model follows these instructions to generate responses that meet the set criteria.

Module 3: Prompt Engineering Techniques for ChatGPT

The Architecture of AI Language Understanding



[Zero-Shot vs Few-Shot Prompting: Complete Guide to AI Prompt Engineering](#)



Module 3: Prompt Engineering Techniques for ChatGPT

Token Analysis

- The input text is divided into small units called tokens (words or sub-words).
- Each token is converted into numerical form using embeddings.
- This allows the model to process language mathematically.

Context Window

- The model considers all tokens within a limited window (its short-term memory).
- It uses self-attention to understand relationships between all tokens in that window.
- Words influence each other based on their position and meaning

Pattern Recognition

- The model identifies patterns and relationships it learned from large training data.
- It uses internal weights to estimate the most likely continuation of text.
- No facts are recalled → only statistical language patterns are used.

Response Generation

- The model predicts one token at a time to form a complete response.
- After each new token, it recalculates context before generating the next one.
- The process repeats until the answer or sentence is complete.

7 – Full Pipeline Summary

Input Text	→ "Hello world"
Tokenization	→ [15496, 2159]
Embeddings	→ Numerical vectors
Attention	→ Relation between tokens
Pattern Search	→ Learned from data
Output Generation	→ "Hello world!"

Process = Encode → Attend → Decode

Module 3: Prompt Engineering Techniques for ChatGPT

1 – Input

User types:
"Hello world"

System receives raw text.
↓
Starts tokenization process.

2 – Tokenization

Input: "Hello world"

Split into tokens:
["Hello", " world"]

Token IDs (example):
[15496, 2159]

3 – Embedding

Each token → vector of real numbers

Example (8-dimensional preview):

"Hello" → [0.21, -0.34, 0.57, 0.89, -0.12, 0.45, 0.03, -0.67]
" world" → [0.11, 0.72, -0.44, 0.36, -0.25, 0.90, -0.15, 0.40]

(Actual model uses 768+ dimensions)

4 – Context Window

All token vectors placed in context window.
Self-attention calculates relationships.

Attention weights (simplified):
Hello → world : 0.78
world → Hello : 0.74

5 – Pattern Recognition

Model references learned language patterns.

Finds pattern: "Hello world" → common greeting phrase.
Prepares next-token prediction probabilities.

6 – Response Generation

Predicted next tokens with probabilities:

"!" → 0.85
"how" → 0.10
"this" → 0.03
"program" → 0.02

Highest probability → "!"

Output generated token by token:
"Hello world!"

Module 3: Prompt Engineering Techniques for ChatGPT



3.Zero,One,and Few Shot Prompting

- Zero, One, and Few Shot Prompting are techniques in prompt engineering that help guide AI models, like ChatGPT, to generate responses based on different amounts of context or examples. These techniques are useful when you want the model to produce responses with limited information.
 - a.Zero Shot Prompting
 - b.One Shot Prompting
 - c.Few Shot Prompting

Module 3: Prompt Engineering Techniques for ChatGPT

3. Zero, One, and Few Shot Prompting

1. Zero Shot Prompting

- The model generates a response without any examples or context. You simply give a prompt or instruction, and the model uses its pre-existing knowledge to generate an answer.
- **Example:**
 - **Prompt:** "Write a poem about nature." **Output:** A poem generated by the
 - model based on its general knowledge.
- **When to Use:**
 - When you want the model to think creatively without any specific examples. Useful for open-ended tasks or when you want to explore the model's generalization abilities.
- **Limitation:**
 - Sometimes results may be less accurate or relevant without context.

Module 3: Prompt Engineering Techniques for ChatGPT

3. Zero, One, and Few Shot Prompting

2. One Shot Prompting

- **What It Is:**
- The model generates a response based on one example you provide. This single example helps guide the model in understanding the context or task.
- **Example:**
 - **Prompt:** "Summarize the following paragraph: 'The sun rises in the east and sets in the west.'"
 - **Output:** "The sun moves across the sky from east to west."
- **When to Use:**
 - When you have one clear example but still need the model to generate a response with some level of accuracy.
- **Limitation:**
 - If the task is complex, one example might not be enough, and you may need more examples.

Module 3: Prompt Engineering Techniques for ChatGPT

3. Zero, One, and Few Shot Prompting

3. Few Shot Prompting

- The model generates a response based on a few examples you provide. This gives the model more context and helps it generate more accurate and relevant responses.
- **Example:**
 - **Prompt:** *"Write a product review based on these features:
 - Battery life: 10 hours
 - Camera: 12 MP
 - Screen: 6.5 inches"
 - **Output:** A detailed product review based on the provided features.
- **When to Use:**
 - When you need more context or examples to help the model generate precise and relevant responses.
- **Limitation:**
 - You need to ensure that the examples are relevant and complete to get the best results.

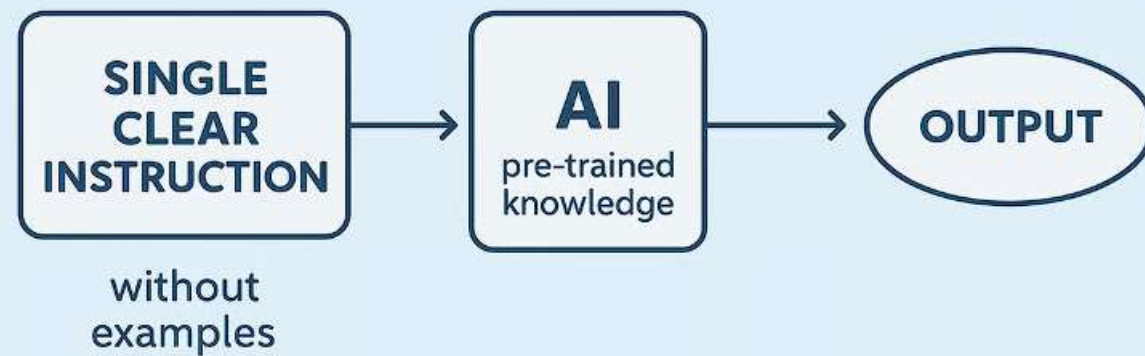
Module 3: Prompt Engineering Techniques for ChatGPT

3. Zero, One, and Few Shot Prompting

Technique	Description	Example
Zero Shot Prompting	The model generates a response without any example or context.	"Write a poem about nature."
One Shot Prompting	The model uses one example to understand the task and generate a response.	"Summarize this sentence: 'The sun rises in the east.'"
Few Shot Prompting	The model uses a few examples to generate a more accurate and relevant response.	"Write a product review based on these features..."

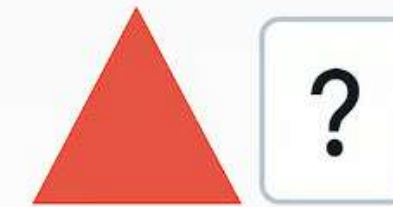
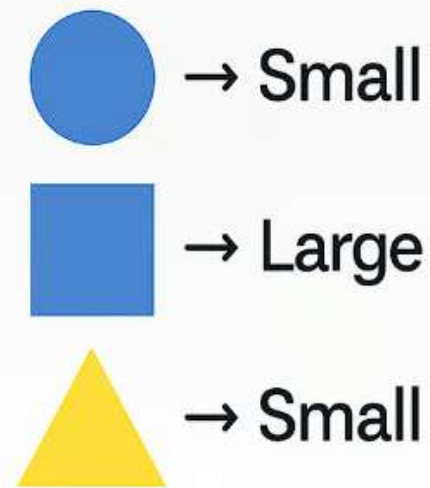
Module 3: Prompt Engineering Techniques for ChatGPT

ZERO-SHOT PROMPTING WORKFLOW

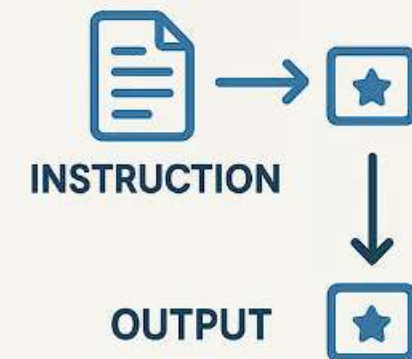


Few-shot prompting example

Pattern-based classification for AI models

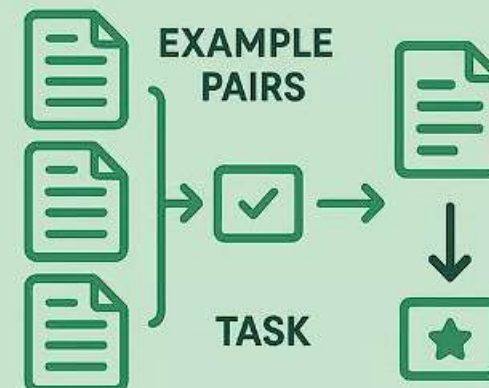


ZERO-SHOT



-  FAST
-  LESS ACCURATE
-  NO SETUP

FEW-SHOT



-  SLOWER
-  MORE ACCURATE
-  WITH SETUP

Module 3: Prompt Engineering Techniques for ChatGPT



4Self-ConsistencyPrompt

- The Self-Consistency Prompt is a sophisticated technique used in prompt engineering to ensure that the generated text remains coherent and logically consistent with the information provided.
- This technique is particularly useful when you want the AI model, such as ChatGPT, to adhere to specific facts, opinions, or constraints throughout a conversation or task.
- **How It Works:**
 - The main aim of the Self-Consistency Prompt is to maintain coherence and logical consistency in the model's responses.
- **Process:**
 - You provide the model with a prompt that contains key information or facts that the model must remain consistent with during the interaction. This could include background knowledge, established facts, or constraints that the model should adhere to while generating text.

Module 3: Prompt Engineering Techniques for ChatGPT

4Self-ConsistencyPrompt



Key Features of Self-Consistency Prompts:

- **Incorporates Key Information:**
 - The prompt includes critical facts, statements, or context that guide the model to maintain consistency.
- **Guides Logical Flow:**
 - Ensures that the model's responses remain logically aligned with the established context throughout the conversation.
- **Balances Detail and Flexibility:**
 - While it's important to provide enough context, it's crucial not to over-restrict the prompt, as that could limit the model's creativity and diversity in responses.

Module 3: Prompt Engineering Techniques for ChatGPT

4Self-ConsistencyPrompt



Example:

1. Topic: Climate Change

- **Prompt:**

- "Climate change is driven by human activity, particularly the burning of fossil fuels. Scientists agree that the Earth's average temperature has risen by 1. °C since the late 19th century. Based on this scientific consensus, explain the impact of rising global temperatures on ecosystems."

- **Generated Response:**

- The model, guided by the self-consistency prompt, would provide an answer that aligns with the scientific consensus and remains consistent with the data about climate change.

Module 3: Prompt Engineering Techniques for ChatGPT

4Self-Consistency Prompt



When to Use Self-Consistency Prompts:

- **Educational Content:**
 - Ensures the information presented is accurate and aligned with established knowledge, making it useful for creating educational materials.
- **Technical Discussions:**
 - When discussing specialized topics, like engineering or medicine, where accuracy and adherence to known facts are critical.
- **Decision-Making:**
 - In scenarios where decisions rely on consistent data or logical reasoning, like business strategies or technical choices

Module 3: Introduction to ChatGPT

ChatGPT is an AI-powered chatbot developed by OpenAI based on the GPT (Generative Pre-trained Transformer) architecture. It is designed to understand and generate human-like text, making it capable of engaging in natural conversations, answering questions, providing explanations, and assisting with a wide range of tasks

GPT stands for Generative Pre-trained Transformer, a type of AI model that:

- Generates text (e.g., stories, emails, summaries)
- Is pre-trained on vast amounts of internet data
- Uses a Transformer architecture, a type of neural network optimized for language tasks

Module 3: Introduction to ChatGPT

ChatGPT can:

- Answer questions and explain concepts
- Write essays, stories, and poems
- Help with coding and debugging
- Translate languages
- Summarize long texts
- Simulate characters or personalities
- Act as a tutor, brainstorming partner, or assistant

ChatGPT was trained on massive datasets from books, articles, websites, and conversations.

It predicts the next word in a sentence, allowing it to generate coherent and relevant responses.

The model does not "know" facts like a human, but it is excellent at mimicking understanding based on patterns in data.

Module 3: History from GPT-1 to GPT-4

GPT-1 (2018)

Paper: *Improving Language Understanding by Generative Pre-Training*

Parameters: 117 million

Architecture: Transformer decoder (12 layers)

Training Objective: Two-phase:

Unsupervised pre-training on Books Corpus (unsupervised language modeling)

Supervised fine-tuning on specific tasks (e.g., question answering, classification)

Significance:

Demonstrated that **unsupervised pre-training** followed by supervised fine-tuning could task-specific models.

Laid the foundation for transfer learning in NLP using generative models.

Module 3: History from GPT-1 to GPT-4

GPT-2 (2019)

Paper: *Language Models are Unsupervised Multitask Learners*

Parameters: 1.5 billion (largest version; smaller variants also released)

Training Data: WebText (~8 million documents scraped from outbound Reddit links)

Architecture: Larger transformer with 48 layers

Key Features:

Trained to predict the next word in a sentence (causal language modeling)

Could perform a range of NLP tasks **zero-shot** (e.g., summarization, translation, question answering) without task-specific training

Controversy:

Initially withheld due to concerns about misuse (e.g., fake news generation)

Eventually fully released

Module 3: History from GPT-1 to GPT-4

GPT-3 (2020)

Paper: *Language Models are Few-Shot Learners*

Parameters: 175 billion

Training Data: Massive dataset including Common Crawl, books, Wikipedia, and more

Key Innovation:

Strong **zero-shot, one-shot, and few-shot learning** capabilities

Could generalize to new tasks with just a few examples in the prompt

Limitations:

No grounding in real-world facts (prone to hallucination)

No memory of past interactions

Bias and ethical concerns

Module 3: History from GPT-1 to GPT-4



GPT-4 (2023)

Paper: *Details limited (OpenAI did not release full technical paper)*

Parameters: Not publicly disclosed (rumored to be in the range of 500B+ or multimodal architecture)

Multimodal Capabilities:

Can process **text and image inputs** (GPT-4 Vision)

Performance:

More **factual, reliable, and steerable** than GPT-3.5

Strong performance on standardized tests (e.g., Bar Exam, SAT)

Module 3: History from GPT-1 to GPT-4



Features:

Better **context handling** (up to 128K tokens for some versions)

Used in tools like **ChatGPT-4 (Pro), Bing Chat, Microsoft Copilot**

Tool use: Can browse the web, write code, generate images (via plugins like DALL·E, Python tools)

Steerability:

Better at adhering to system instructions and user preferences

Introduced “**Custom Instructions**” and memory features in ChatGPT

Module 3: History from GPT-1 to GPT-4



Summary Table

<u>Model</u>	<u>Year</u>	<u>Parameters</u>	<u>Key Innovation</u>	<u>Input Modality</u>
GPT-1	2018	117M	Pre-training + fine-tuning	Text
GPT-2	2019	1.5B	Zero-shot capabilities	Text
GPT-3	2020	175B	Few-shot learning	Text
GPT-4	2023	Undisclosed	Multimodal, better reasoning & memory	Text & Images

Module 3: How to install ChatGPT

Use ChatGPT (No Installation Needed)

The easiest way is to just use ChatGPT directly via your browser or mobile app — no installation required:

◆ Web App

Go to: <https://chat.openai.com>

Sign in or create an OpenAI account.

Start chatting (GPT-3.5 is free; GPT-4 is available with **ChatGPT Plus** subscription).

◆ Mobile Apps (iOS & Android)

You can **install ChatGPT** like any app:

iOS:

Open the **App Store**

Search: "**ChatGPT – OpenAI**"

Download the official app by OpenAI

Module 3: How to install ChatGPT

Android:

Open the **Google Play Store**

Search: "**ChatGPT – OpenAI**"

Install the official app

Use ChatGPT via Microsoft Products

ChatGPT is also built into:

Microsoft Edge browser (via Bing Chat)

Microsoft Word/ Excel as Copilot

No installation is required if you're using updated Microsoft software

Module 3: How to install ChatGPT

Install/OpenAI API Locally (Developers Only)

If you're a developer and want to **build your own app using ChatGPT**, you can use the **OpenAI API**.

Requirements:

A programming environment (Python, Node.js, etc.)

OpenAI API key (get it from <https://platform.openai.com>)

What You *Can't* Install:

You **can't install GPT-4 locally** (it runs on powerful servers in the cloud).

OpenAI has **not released GPT-4 weights** (not open-source).

Local models like **LLaMA, Mistral, or GPT4All** can be installed instead, but they are not the same as ChatGPT.

Module 3: How to Configure ChatGPT

Configuring ChatGPT (Web or Mobile App)

If you're using chat.openai.com or the mobile app:

Use Custom Instructions (Web & App)

This lets you customize how ChatGPT responds to you.

🔗 **How to set it up:**

Go to ChatGPT (web or mobile)

Click your name or the three-dot menu

Choose "Custom Instructions"

Fill in:

What would you like ChatGPT to know about you?

How would you like ChatGPT to respond?

Save

Module 3: How to Configure ChatGPT

You can tell it:

Preferred tone (e.g., casual, formal)

Areas of focus (e.g., tech, fitness)

Specific writing formats (e.g., Markdown, bullet points)

Enable/Disable Memory (Web Only, for now)

This lets ChatGPT **remember facts** about you across chats.

To check or configure:

Go to chat.openai.com

Click your name > **Settings**

Go to **Personalization > Memory**

Toggle memory on/off and manage stored information

Module 3: How to Configure ChatGPT

Configuring ChatGPT via the API (Developers)

If you're using the **OpenAI API**, configuration happens through code

⚙️ Key Config Options:

- model: Choose "gpt-3.5-turbo" or "gpt-4"
- temperature: Controls randomness (0 = deterministic, 1 = creative)
- max_tokens: Response length limit
- system message: Define assistant behavior or role
- tools: Add tool use (e.g., functions, code interpreter)

Module 3: How to Configure ChatGPT



Using ChatGPT in Microsoft Copilot/ Bing Chat

These versions use ChatGPT under the hood, but **you can't directly configure** it like in the OpenAI platform.

However, you can **influence behavior with good prompts**, such as:

"Explain this as if I'm a beginner."

"Give me a summary in bullet points."

"Use a professional tone."

Thank You Module 3 -

Completed